

CS601 FINAL TERM SOLVED MCQs

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Question No: 1 (Marks: 1) - Please choose one

Which is not an element of protocol

- ▶ semantics
- ▶ timing
- ▶ [communication service module](#)

Question No: 2 (Marks: 1) - Please choose one

Layers 5, 6 and 7 also called as network support layers.

- ▶ [True](#)
- ▶ False

Question No: 3 (Marks: 1) - Please choose one

Fourier transform tells us that any digital signal can be decomposed into infinite number of periodic signals

- ▶ [True](#)
- ▶ False

Question No: 4 (Marks: 1) - Please choose one

Time domain plot show changes in signal phase with respect to time.

- ▶ True
- ▶ False

Question No: 5 (Marks: 1) - Please choose one

Analog to digital conversion is also termed as modulating an analog signal.

- ▶ True
- ▶ False

Question No: 6 (Marks: 1) - Please choose one

DC component is also termed as Direct current component or a component with non-zero frequency.

- ▶ False
- ▶ True

Question No: 7 (Marks: 1) - Please choose one

Manchester is a type of _____encoding.

- ▶ biphase
- ▶ polar

- ▶ biphase & polar
- ▶ none of the given

Question No: 8 (Marks: 1) - Please choose one

The inversion of the level at 1 bit is called as _____

- ▶ NRZ-L
- ▶ NRZ-I
- ▶ RZ

Question No: 9 (Marks: 1) - Please choose one

PCM is the first process of PAM.

- ▶ True
- ▶ False

Question No: 10 (Marks: 1) - Please choose one

In 4PSK each phase change represents _____bits.

- ▶ 3
- ▶ 6
- ▶ 2
- ▶ 4

Question No: 11 (Marks: 1) - Please choose one

In RS 422 Balanced mode two lines carry _____ signals which are not identical to each other.

- ▶ same
- ▶ different
- ▶ digital
- ▶ analog

Question No: 12 (Marks: 1) - Please choose one

A _____ converts an analog signal into a digital signal.

- ▶ Demodulator
- ▶ Modulator
- ▶ Digital-to-analog converter

Question No: 13 (Marks: 1) - Please choose one

Which of the following is an example of ITU-T modem standards:

- ▶ T-series
- ▶ X-series
- ▶ N-series
- ▶ V-series

Question No: 14 (Marks: 1) - Please choose one

The maximum data rate in the uploading direction is still _____.

- ▶ 26.6 Kbps
- ▶ 33.6 Kbps
- ▶ 36.6 Kbps
- ▶ 46.6 Kbps

Question No: 15 (Marks: 1) - Please choose one

Which of the following primarily uses guided media

- ▶ Cellular telephone system
- ▶ Local telephone system
- ▶ Satellite communications
- ▶ Radio broadcasting

Question No: 16 (Marks: 1) - Please choose one

When a beam of light travels through media of two different densities, if the angle of incidence is greater than the critical angle, _____ occurs.

- ▶ Reflection
- ▶ Refraction
- ▶ Incidence
- ▶ Criticism

Question No: 17 (Marks: 1) - Please choose one

When we talk about unguided media, usually we are referring to _____.

- ▶ Metallic wires
- ▶ Nonmetallic wires
- ▶ The air
- ▶ Water

Question No: 18 (Marks: 1) - Please choose one

Optical fibers are defined by the ratio of the _____ of their core to the diameter of their cladding.

- ▶ Diameter
- ▶ Radius
- ▶ Length
- ▶ Width

Question No: 19 (Marks: 1) - Please choose one

All of popular Fiber optic connectors are _____ shaped.

- ▶ Conical
- ▶ Barrel
- ▶ Circular
- ▶ Rectangular

Question No: 20 (Marks: 1) - Please choose one

Radio wave transmission utilizes _____ different types of propagation.

- ▶ Four
- ▶ Three
- ▶ Two
- ▶ Five

Question No: 21 (Marks: 1) - Please choose one

dB is _____ if a signal is amplified.

- ▶ Negative
- ▶ Positive
- ▶ Null
- ▶ Zero

Question No: 22 (Marks: 1) - Please choose one

A prism can deflect the light depending upon the angle of _____ and the frequency.

- ▶ Deviation
- ▶ Incident
- ▶ Refraction
- ▶ Reflection

Question No: 23 (Marks: 1) - Please choose one

Asynchronous TDM is efficient only when the size of the time slot is kept relatively _____

- ▶ Large
- ▶ Small
- ▶ Medium
- ▶ None of the given

Question No: 24 (Marks: 1) - Please choose one

The local loop has _____ cable that connects the subscriber telephone to the nearest end office.

- ▶ Twisted-pair
- ▶ Coaxial
- ▶ Fiber-optic
- ▶ None of the given

Question No: 25 (Marks: 1) - Please choose one

Data from a computer are _____; the local loop handles _____ signals.

- ▶ Analog; analog
- ▶ Analog; digital
- ▶ Digital; digital
- ▶ Digital; analog

Question No: 26 (Marks: 1) - Please choose one

Which error detection method uses ones complement arithmetic?

- ▶ Simple parity check
- ▶ Two-dimensional parity check
- ▶ CRC
- ▶ Checksum

Question No: 27 (Marks: 1) - Please choose one

Flow control is needed to prevent _____

- ▶ Bit errors
- ▶ Overflow of the sender buffer
- ▶ Overflow of the receiver buffer
- ▶ Collision between sender and receiver

Question No: 28 (Marks: 1) - Please choose one

_____ coordinates the amount of data that can be sent before receiving acknowledgment

- ▶ flow control
- ▶ error control
- ▶ data control

Question No: 29 (Marks: 1) - Please choose one

Error control is both error _____ and error _____

- ▶ detection; correction
- ▶ detection; deletion
- ▶ detection; avoidance
- ▶ detection; forwarding

Question No: 30 (Marks: 1) - Please choose one

Addressing is not needed in _____ configuration.

- ▶ Point to Point
- ▶ Multipoint
- ▶ Point to point and multipoint

Question No: 31 (Marks: 1) - Please choose one

In a Go-Back-N ARQ, if the window size is 63, what is the range of sequence number?

- ▶ 0 to 63
- ▶ 0 to 64
- ▶ 1 to 63
- ▶ 1 to 64

Question No: 32 (Marks: 1) - Please choose one

A timer is set when _____ is sent out.

- ▶ A data frame
- ▶ An ACK
- ▶ A NAK

Question No: 33 (Marks: 1) - Please choose one

Sliding window requires that data frames be transmitted _____

- ▶ Sequentially
- ▶ Frequently
- ▶ Synchronously
- ▶ Asynchronously

Question No: 34 (Marks: 1) - Please choose one

In Y-MODEM Multiple files can be sent simultaneously

- ▶ True
- ▶ False

Question No: 35 (Marks: 1) - Please choose one

BLAST stands for :

- ▶ Blocked asynchronous transmission
- ▶ Blocked synchronous transmission

- ▶ Barrel asynchronous transmission
- ▶ Below asynchronous transmission

Question No: 36 (Marks: 1) - Please choose one

HDLC is an acronym for _____.

- ▶ High-duplex line communication
- ▶ High-level data link control
- ▶ Half-duplex digital link combination
- ▶ Host double-level circuit

Question No: 37 (Marks: 1) - Please choose one

The HDLC _____ field defines the beginning and end of a frame.

- ▶ Flag
- ▶ Address
- ▶ Control
- ▶ FCS

Question No: 38 (Marks: 1) - Please choose one

_____ is the access protocol used by traditional Ethernet.

- ▶ CSMA/CD
- ▶ CSMA/CA

▶ Token Ring

▶ CSMA

Question No: 39 (Marks: 1) - Please choose one

Bridges can divide a large _____ into smaller segments

▶ Network

▶ Packet

▶ Frame

▶ Address

Question No: 40 (Marks: 1) - Please choose one

Like VRC, LRC and CRC, Checksum is also based on _____.

▶ Redundency

▶ Decimal Division

▶ Encryption

▶ Encoding

Question No: 41 (Marks: 2)

What are the conditions for the polynomial used by the CRC generator?

CRC generator:

CRC generator (the divisor) is most often represented not as a 1's and 0's but as an algebraic polynomial.

conditions for the polynomial:

it should have following properties:

It should not be divisible by "x".

It should not be divisible by "x+1".

The first condition guarantees that all burst error of a length equal to degree of the polynomial is detected.

The 2nd condition guarantees that all burst error affecting an odd number of bits are detected.

Question No: 42 (Marks: 2)

What are intelligent modems?

Intelligent modems:

A modem that responds to commands and can accept new instructions during online transmission. It was originally developed by Hayes.

Example:

- Automatic answering,
- Dialing etc.

Question No: 43 (Marks: 2)

What is the basic purpose of Router?

Basic purpose of Router:

"A router is a device that extracts the destination of a packet it receives, selects the best path to that destination, and forwards data packets to the next device along this path. They connect networks together;

a LAN to a WAN for example, to access the Internet.

"A more precise definition of a router is a computer networking device that interconnects separate logical subnets."

Question No: 44 (Marks: 3)

What are the fractional T Lines?

The fractional T Lines:

Many subscribers don't need the entire capacity of the T-line.

For example,

A small business may need only one-fourth of the capacity of T-line. if four business of same size lie in the same building, they can share T-line. DSU/CSU allow the capacity of T-line to be interleaved in to four channels

Question No: 45 (Marks: 3)

What are the light sources used for optic fiber?

- **light sources used for optic fiber:**
- The light source can be an LED or ILD
- **LED** (Light emitting diode) cheaper but provide unfocused light that strikes the boundaries of channel at uncontrollable angles.
- Limited to short distance use.
- **LASSER**
- Can be focused to a narrow range allowing control over angle of incidence.

Question No: 46 (Marks: 3)

What is Multi Access Unit (MAU) in Token Ring?

Multi Access Unit (MAU) in Token Ring:

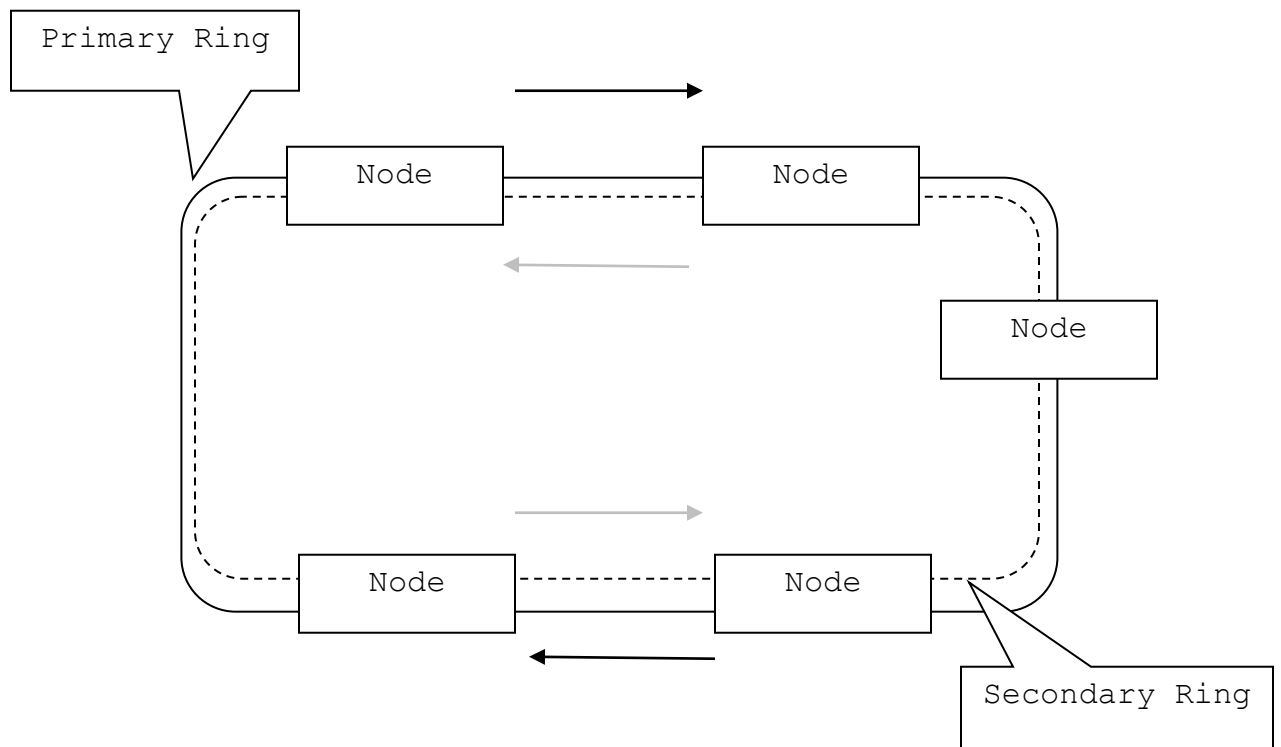
- Individual automatic switches are combined in to a hub
- One MAU can support up to 8 stations.
- Although it looks like a star, it is in fact a ring.

Question No: 47 (Marks: 5)

Give characteristics of Dual Ring, if necessary then draw the diagram. [5]

Characteristics of Dual Ring:

A network topology in which two concentric rings connect each node on a network instead of one network ring that is used in a ring topology. Typically, the secondary ring in a dual-ring topology is redundant. It is used as a backup in case the primary ring fails. In these configurations, data moves in opposite directions around the rings. Each ring is independent of the other until the primary ring fails and the two rings are connected to continue the flow of data traffic.



What the receiver will receive if the checksum method is applied to the following bit.
10101001 00111001

Ans:

the receiver will receive the checksum method is applied to the following bit.

10101001 00111001

10101001 00111001

Sum of 2 bits are

10101001

00111001

11100010

00011101 1's complement

1

00011110 2's complement

10101001 00111001 ==> 00011110

So the data transmitted which will receiver get:

10101001 00111001 00011110

Question No: 49 (Marks: 5)

What is refraction in terms of optic fiber? Give one example.

Refraction:

Light travels in a straight line as long as it is moving through a single uniform

structure If a ray of light traveling through one substance enters another (more or less dense) substance, its speed changes abruptly causing the ray to change direction. This phenomenon is called Refraction.

Refraction in terms of optic fiber:

the propagation of light in an optical fiber which in its simplest form consists of a circular core of uniform refractive index surrounded by a cladding of slightly lower refractive index. The light is launched into the entrance face of the fiber.

The light is propagated by the total internal reflection at the interface between core and cladding. However the rays incident at angles larger than a certain angle, called the cut-off angle, suffer both refraction and reflection at the interface between the core and the cladding.

They, therefore, are not guided. Due to this the optical fiber has a numerical aperture. The numerical aperture is given by the square root of $(n_1^2 - n_2^2)$. Typical values of numerical aperture lie between 0.1 and 0.3.

The refractive indices of the core and the cladding are n_1 and n_2 respectively. The fiber is normally in air ($n_0=1$) but could also be in a medium of refractive index n_0 .

Question No: 50 (Marks: 10)

What are the asynchronous protocols in data link layer? Discuss in detail with examples. [10 marks]

Asynchronous protocols in data link layer:

Asynchronous communication at the data link layer or higher protocol layers is known as statistical multiplexing or packet mode communication,

For example :

Asynchronous transfer mode (ATM). In this case the asynchronously transferred blocks are called data packets,

Async protocols in Data link layer is called statistical multiplexing. for example ATM cells.

The opposite is circuit switched communication, which provides constant bit rate, for example ISDN and SONET/SDH.

The packets may be encapsulated in a data frame, with a frame synchronization bit sequence indicating the start of the frame, and sometimes also a bit

synchronization bit sequence, typically 01010101, for identification of the bit transition times. Note that at the physical layer, this is considered as synchronous serial communication.

Examples of packet mode data link protocols that can be/are transferred using synchronous serial communication are the

- HDLC,
- Ethernet,
- PPP and
- USB protocols.

(GOOD LUCK FOR FINAL EXAM)

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